
Chapter 2: Instruction Set Architecture

(Language of the Computer)

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[with materials from *Computer Organization and Design, 4th Edition*,
Patterson & Hennessy, © 2008, MK
and M.J. Irwin's presentation, PSU 2008]

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- ❑ MIPS Instruction Set Architecture
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What is MIPS, and why MIPS?

- ❑ CPU designed by John Hennessy's team
 - | Stanford's president 2000-2016
 - | 2017 Turing award for RISC development
 - | “god father” of Silicon Valley
- ❑ Very successful CPU in 80s-90s, the first that have 64 bit architecture
- ❑ Still very popular in embedded market: set top box, game console,...
- ❑ Simple instruction set, appropriate for education (the mini instruction set)

Computer language: hardware operation

❑ Want to command the computer?

➔ You need to speak its language!!!

❑ Example: MIPS assembly instruction

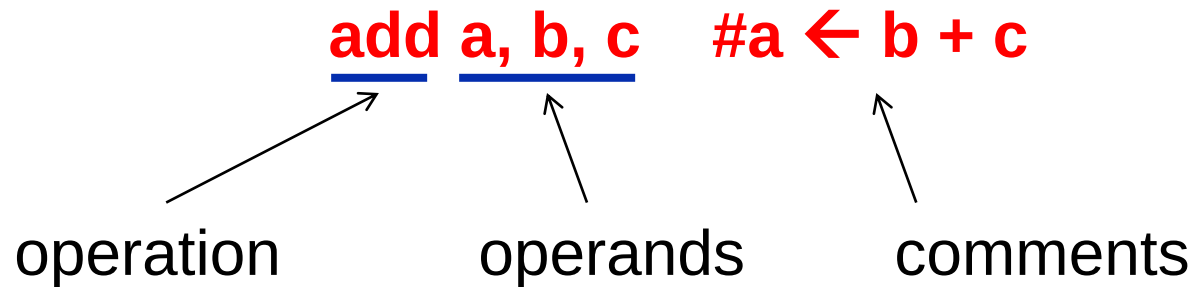
add a, b, c #a ← b + c

❑ Operation performed

- ❑ add b and c,
- ❑ then store result into a

add a, b, c #a ← b + c

operation operands comments



Hardware operation

- ❑ What does the following code do?

```
add t0, g, h  
add t1, i, j  
sub f, t0, t1
```

- ❑ Equivalent C code

$$f = (g + h) - (i + j)$$

➔ *Why not making 4 or 5 inputs instructions?*

➔ ***DP1: Simplicity favors regularity!***

Operands

❑ Object of operation

- | Source operand: provides input data
- | Destination operand: stores the result of operation

❑ MIPS operands

- | Registers
- | Memory locations
- | Constant/Immediate

MIPS operands

Name	Example	Comments
32 registers	<code>\$s0-\$s7</code> , <code>\$t0-\$t9</code> , <code>\$zero</code> , <code>\$a0-\$a3</code> , <code>\$v0-\$v1</code> , <code>\$gp</code> , <code>\$fp</code> , <code>\$sp</code> , <code>\$ra</code> , <code>\$at</code>	Fast locations for data. In MIPS, data must be in registers to perform arithmetic, register <code>\$zero</code> always equals 0, and register <code>\$at</code> is reserved by the assembler to handle large constants.
2^{30} memory words	<code>Memory[0]</code> , <code>Memory[4]</code> , . . . , <code>Memory[4294967292]</code>	Accessed only by data transfer instructions. MIPS uses byte addresses, so sequential word addresses differ by 4. Memory holds data structures, arrays, and spilled registers.

Register operand: MIPS Register File

- ❑ Special memory inside CPU, called register file
- ❑ 32 slots, each slot is called a register
- ❑ Each register holds 32 bits of data (a word)
- ❑ Each register has a unique address, and a name
- ❑ Register's address is from 0 to 31, represented by 5 bits

Data types in MIPS

Byte = 8 bits



Halfword = 2 bytes



Word = 4 bytes



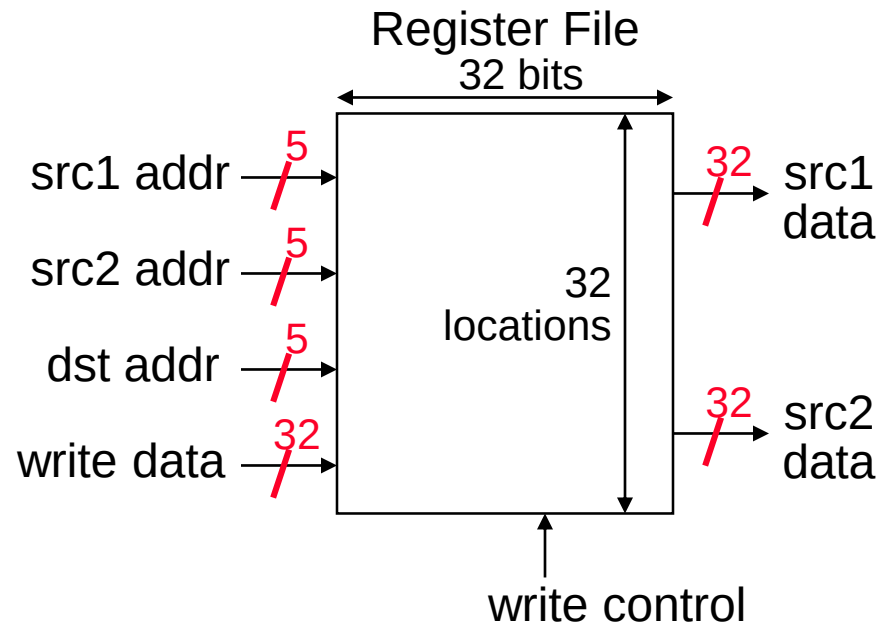
Doubleword = 8 bytes



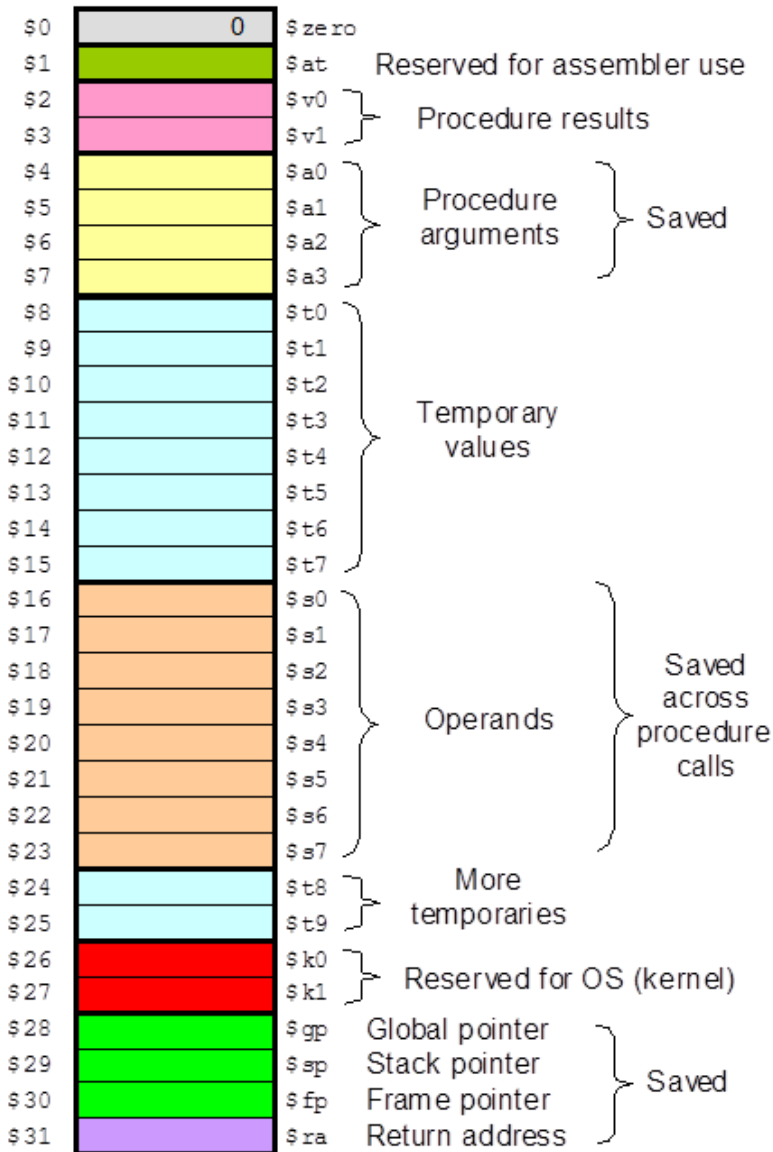
MIPS32 registers hold 32-bit (4-byte) words. Other common data sizes include byte, halfword, and doubleword.

Register operand: MIPS Register File

- ❑ Register file in MIPS CPU
 - | Two read ports with two source address
 - | One write port with one destination address
 - | Located in CPU → fast, small size



MIPS Register Convention



❑ MIPS: load/store machine.

❑ Typical operation

- | Load data from memory to register
- | **Data processing in CPU**
- | Store data from register to memory

Register operand: MIPS Register File

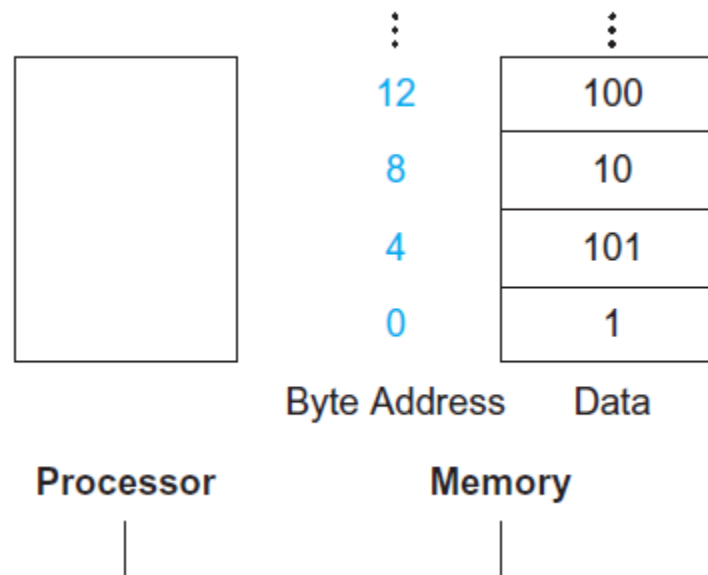
- ❑ Register file: “work place” right inside CPU.
- ❑ Larger register file should be better, more flexibility for CPU operation.
- ❑ Moore’s law: doubled number of transistor every 18 mo.
- ❑ Why only 32 registers, not more?

➔ DP2: *Smaller is faster!*

Effective use of register file is critical!

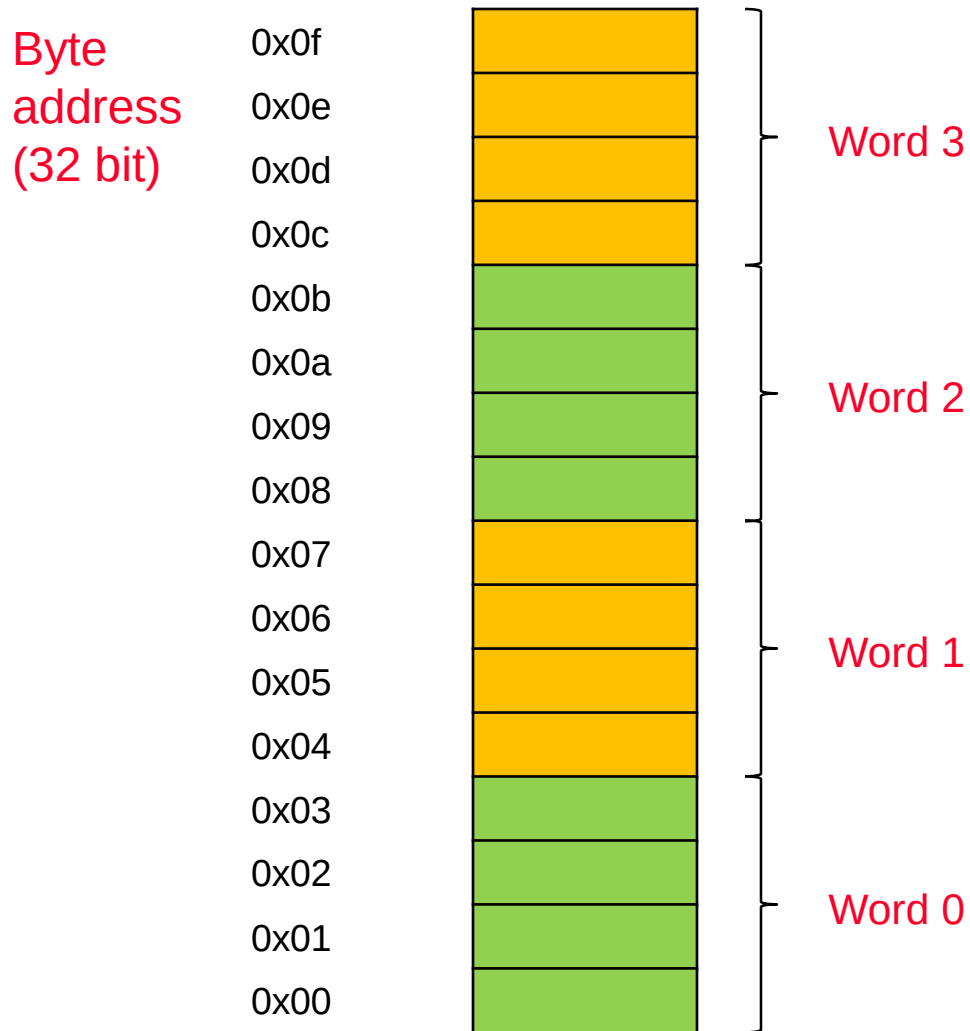
Memory operand

- ❑ Data stored in computer's main memory
 - | Large size
 - | Outsize CPU → Slower than register
- ❑ Operations with memory operand
 - | Load values from memory to register
 - | Store result from register to memory



Byte addressable
Word aligned

MIPS memory organization



- ❑ Byte addressable
- ❑ Word data access via byte address
- ❑ **Only accessible via load/store instructions**

Alignment

In decimal:

$$\text{Word address} = 4 * \text{word number}$$

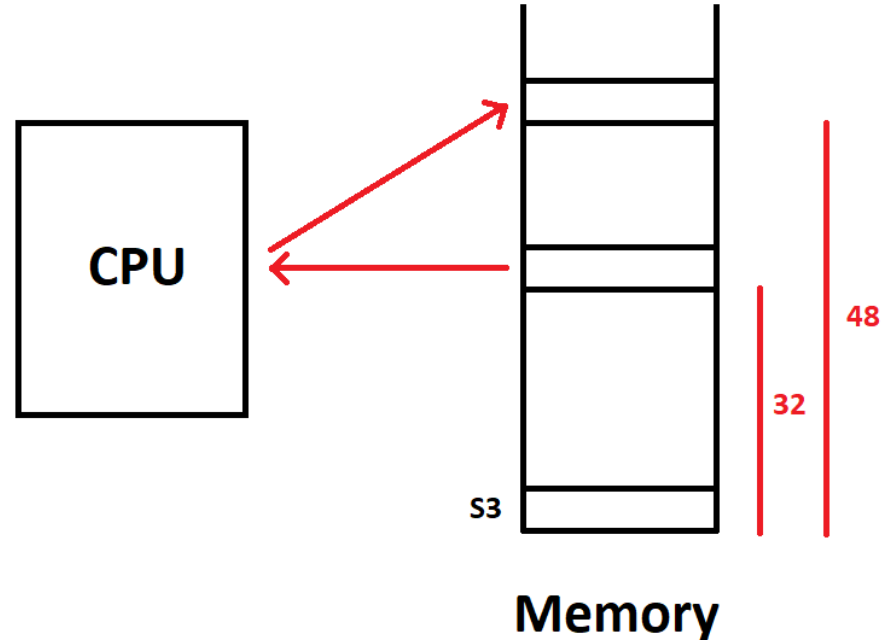
In binary:

$$\text{Word address} = \text{word number} + 00$$

Memory operand

❑ Sample instruction

```
lw $t0, 32($s3)
#do sth
#
sw $t0, 48($s3)
```



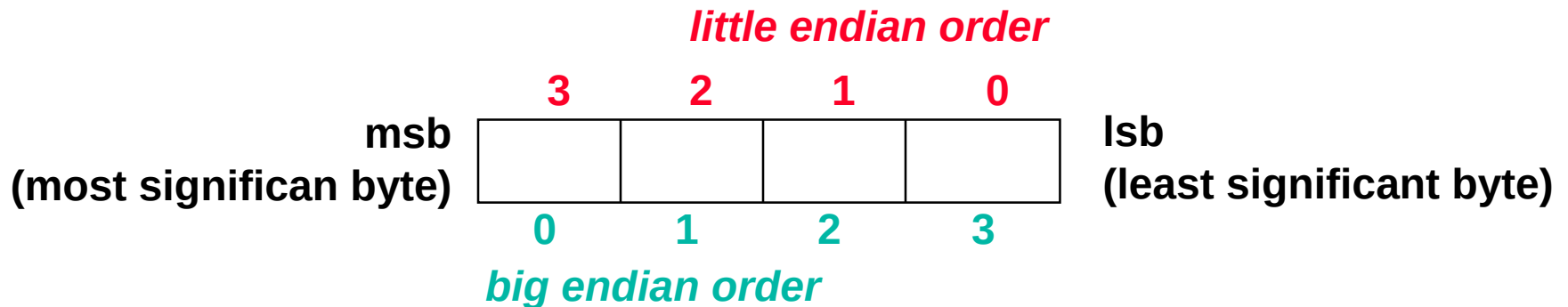
Byte Addresses

❑ **Big Endian:** leftmost byte is word address

IBM 360/370, Motorola 68k, **MIPS**, Sparc, HP PA

❑ **Little Endian:** rightmost byte is word address

Intel 80x86, DEC Vax, DEC Alpha (Windows NT)



Example

- ❑ Consider a word in MIPS memory consists of 4 byte with hexa value as below
- ❑ What is the word's value?

address	value
X+3	68
X+2	1B
X+1	5D
X	FA

- ❑ MIPS is big-endian: address of MSB is X
- ➔ word's value: FA5D1B68

Immediate operand

- ❑ Immediate value specified by the constant number
- ❑ Does not need to be stored in register file or memory
 - | Value encoded right in instruction → very fast
 - | Fixed value specified when developing the program
 - | Cannot change value at run time

Immediate operand

- ❑ What is the mostly used constant?
- ❑ The special register: \$zero
- ❑ Constant value of 0
- ❑ Why?

➔ *DP3: Making common cases fast!*

What are stored inside operands?

- ❑ Data, of course!
- ❑ And data is represented as binary numbers
- ❑ Then how binary numbers are treated by MIPS?
 - | As integers
 - | Unsigned
 - | Signed

Unsigned Binary Integers

- ❑ Using n-bit binary number to represent non-negative integer

$$\begin{aligned} X &= X_{n-1}X_{n-2}\dots X_1X_0 \\ &= X_{n-1}2^{n-1} + X_{n-2}2^{n-2} + \dots + X_12^1 + X_02^0 \end{aligned}$$

- ❑ Range: 0 to $+2^n - 1$

- ❑ Example

$$\begin{aligned} &0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 1011_2 \\ &= 0 + \dots + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 0 + \dots + 8 + 0 + 2 + 1 = 11_{10} \end{aligned}$$

- ❑ Data range using 32 bits

$$0 \text{ to } 2^{32}-1 = 4,294,967,295$$

Eg: 32 bit Unsigned Binary Integers

Hex	Binary	Decimal
0x00000000	0...0000	0
0x00000001	0...0001	1
0x00000002	0...0010	2
0x00000003	0...0011	3
0x00000004	0...0100	4
0x00000005	0...0101	5
0x00000006	0...0110	6
0x00000007	0...0111	7
0x00000008	0...1000	8
0x00000009	0...1001	9
	...	
0xFFFFFFF0	1...1100	$2^{32}-4$
0xFFFFFFF1	1...1101	$2^{32}-3$
0xFFFFFFF2	1...1110	$2^{32}-2$
0xFFFFFFFF	1...1111	$2^{32}-1$

Exercise

❑ Convert to 32 bit integers

25 = 0000 0000 0000 0000 0000 0000 0001 1001

125 = 0000 0000 0000 0000 0000 0000 0111 1101

255 = 0000 0000 0000 0000 0000 0000 1111 1111

❑ Convert 32 bit integers to decimal value

0000 0000 0000 0000 0000 0000 1100 1111 = 207

0000 0000 0000 0000 0000 0001 0011 0011 = 307

Signed binary integers

- Using n-bit binary number to represent integer, including negative values

$$\begin{aligned} X &= X_{n-1}X_{n-2}\dots X_1X_0 \\ &= -X_{n-1}2^{n-1} + X_{n-2}2^{n-2} + \dots + X_12^1 + X_02^0 \end{aligned}$$

- Range: -2^{n-1} to $+2^{n-1} - 1$

- Example

$$\begin{aligned} &1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1100_2 \\ &= -1 \times 2^{31} + 1 \times 2^{30} + \dots + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 \\ &= -2,147,483,648 + 2,147,483,644 = -4_{10} \end{aligned}$$

- Using 32 bits

$$-2,147,483,648 \text{ to } +2,147,483,647$$

Signed integer negation

- ❑ Given $x = x_{n-1}x_{n-2} \dots x_1x_0$, how to calculate $-x$?
- ❑ Let $\bar{x}$ = 1's complement of x

$$\bar{x} = 1111 \dots 11_2 - x$$

$$(1 \rightarrow 0, 0 \rightarrow 1)$$

Then

$$\bar{x} + x = 1111 \dots 11_2 = -1$$

$$\Rightarrow \bar{x} + 1 = -x$$

- ❑ Example: find binary representation of -2

$$+2 = 0000 \ 0000 \dots 0010_2$$

$$\begin{aligned} -2 &= 1111 \ 1111 \dots 1101_2 + 1 \\ &= 1111 \ 1111 \dots 1110_2 \end{aligned}$$

Signed binary negation

$$-2^3 =$$

$$-(2^3 - 1) =$$

2'sc binary	decimal
1000	-8
1001	-7
1010	-6
1011	-5
1100	-4
1101	-3
1110	-2
1111	-1
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7

complement all the bits

0101

1011

and add a 1

and add a 1

0110

1010

complement all the bits

$$2^3 - 1 =$$

Exercise

❑ Find 16 bit signed integer representation of

$$16 = 0000\ 0000\ 0001\ 0000$$

$$-16 = 1111\ 1111\ 1111\ 0000$$

$$100 = 0000\ 0000\ 0110\ 0100$$

$$-100 = 1111\ 1111\ 1001\ 1100$$

Sign extension

- ❑ Given n-bit integer $x = x_{n-1}x_{n-2} \dots x_1x_0$
- ❑ Find corresponding m-bit representation ($m > n$) with the same numeric value

$$x = x_{m-1}x_{m-2} \dots x_1x_0$$

- ❑ → Replicate the sign bit to the left

- ❑ Examples: 8-bit to 16-bit

+2: 0000 0010 $\Rightarrow$ 0000 0000 0000 0010

-2: 1111 1110 $\Rightarrow$ 1111 1111 1111 1110

Instruction set

- ❑ 3 instruction formats:

- | Register (R)
- | Immediate (I)
- | Branch (J)

- ❑ R-instruction: all operands are register

- ❑ I-instruction: one operand is immediate

- ❑ J-instruction: the unconditional branch

- ❑ **Note: All MIPS instructions are 32 bits long**

➔ *Why not only one format?*

➔ *DP4: Good design demands good compromises!*

5 instruction types

- ❑ Arithmetic: addition, subtraction
- ❑ Data transfer: transfer data between registers, memory, and immediate
- ❑ Logical: and, or, shift
- ❑ Conditional branch
- ❑ Unconditional branch

Overview of MIPS instruction set

MIPS assembly language

Category	Instruction	Example	Meaning	Comments
Arithmetic	add	add \$s1,\$s2,\$s3	$\$s1 = \$s2 + \$s3$	Three register operands
	subtract	sub \$s1,\$s2,\$s3	$\$s1 = \$s2 - \$s3$	Three register operands
	add immediate	addi \$s1,\$s2,20	$\$s1 = \$s2 + 20$	Used to add constants
Data transfer	load word	lw \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Word from memory to register
	store word	sw \$s1,20(\$s2)	$\text{Memory}[\$s2 + 20] = \$s1$	Word from register to memory
	load half	lh \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Halfword memory to register
	load half unsigned	lhu \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Halfword memory to register
	store half	sh \$s1,20(\$s2)	$\text{Memory}[\$s2 + 20] = \$s1$	Halfword register to memory
	load byte	lb \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Byte from memory to register
	load byte unsigned	lbu \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Byte from memory to register
	store byte	sb \$s1,20(\$s2)	$\text{Memory}[\$s2 + 20] = \$s1$	Byte from register to memory
	load linked word	ll \$s1,20(\$s2)	$\$s1 = \text{Memory}[\$s2 + 20]$	Load word as 1st half of atomic swap
	store condition. word	sc \$s1,20(\$s2)	$\text{Memory}[\$s2 + 20] = \$s1; \$s1 = 0 \text{ or } 1$	Store word as 2nd half of atomic swap
Logical	load upper immed.	lui \$s1,20	$\$s1 = 20 * 2^{16}$	Loads constant in upper 16 bits
	and	and \$s1,\$s2,\$s3	$\$s1 = \$s2 \& \$s3$	Three reg. operands; bit-by-bit AND
	or	or \$s1,\$s2,\$s3	$\$s1 = \$s2 \mid \$s3$	Three reg. operands; bit-by-bit OR
	nor	nor \$s1,\$s2,\$s3	$\$s1 = \sim (\$s2 \mid \$s3)$	Three reg. operands; bit-by-bit NOR
	and immediate	andi \$s1,\$s2,20	$\$s1 = \$s2 \& 20$	Bit-by-bit AND reg with constant
	or immediate	ori \$s1,\$s2,20	$\$s1 = \$s2 \mid 20$	Bit-by-bit OR reg with constant
	shift left logical	sll \$s1,\$s2,10	$\$s1 = \$s2 \ll 10$	Shift left by constant
Conditional branch	shift right logical	srl \$s1,\$s2,10	$\$s1 = \$s2 \gg 10$	Shift right by constant
	branch on equal	beq \$s1,\$s2,25	if ($\$s1 == \$s2$) go to PC + 4 + 100	Equal test; PC-relative branch
	branch on not equal	bne \$s1,\$s2,25	if ($\$s1 \neq \$s2$) go to PC + 4 + 100	Not equal test; PC-relative
	set on less than	slt \$s1,\$s2,\$s3	if ($\$s2 < \$s3$) $\$s1 = 1$; else $\$s1 = 0$	Compare less than; for beq, bne
	set on less than unsigned	sltu \$s1,\$s2,\$s3	if ($\$s2 < \$s3$) $\$s1 = 1$; else $\$s1 = 0$	Compare less than unsigned
	set less than immediate	slti \$s1,\$s2,20	if ($\$s2 < 20$) $\$s1 = 1$; else $\$s1 = 0$	Compare less than constant
Unconditional jump	set less than immediate unsigned	sltiu \$s1,\$s2,20	if ($\$s2 < 20$) $\$s1 = 1$; else $\$s1 = 0$	Compare less than constant unsigned
	jump	j 2500	go to 10000	Jump to target address
	jump register	jr \$ra	go to \$ra	For switch, procedure return
	jump and link	jal 2500	$\$ra = \text{PC} + 4$; go to 10000	For procedure call

Fig. 2.1

MIPS Instruction set: Arithmetic operations

❑ MIPS arithmetic statement

add rd, rs, rt #rd $\leftarrow$ rs + rt

sub rd, rs, rt #rd $\leftarrow$ rs - rt

addi rd, rs, const #rd $\leftarrow$ rs + const

- rs 5-bits register file address of the first source operand
- rt 5-bits register file address of the second source operand
- rd 5-bits register file address of the result's destination

Example

- ❑ Currently $\$s1 = 6$
- ❑ What is value of $\$s1$ after executing the following instruction

addi $\$s2, \$s1, 3$

addi $\$s1, \$s1, -2$

sub $\$s1, \$s2, \$s1$

MIPS Instruction set: Logical operations

❑ Basic logic operations

`and rd, rs, rt` $\#rd \leftarrow rs \ \& \ rt$

`andi rd, rs, const` $\#rd \leftarrow rs \ \& \ const$

`or rd, rs, rt` $\#rd \leftarrow rs \ | \ rt$

`ori rd, rs, const` $\#rd \leftarrow rs \ | \ const$

`nor rd, rs, rt` $\#rd \leftarrow \sim(rs \ | \ rt)$

❑ Example $\$s1 = 8 = 0000 \ 1000$, $\$s2 = 14 = 0000 \ 1110$

`and $s3, $s1, $s2`

`or $s4, $s1, $s2`

MIPS Instruction set: Logical operations

- ❑ Logical shift and arithmetic shift: move all the bits left or right

sll rd, rs, const #rd $\leftarrow$ rs $\ll$ const

srl rd, rs, const #rd $\leftarrow$ rs $\gg$ const

sra rd, rs, const #rd $\leftarrow$ rs $\gg$ const
(keep sign bit)

MIPS Instruction set: Memory Access Instructions

- ❑ MIPS has two basic **data transfer** instructions for accessing memory

```
lw $t0, 4($s3)    #load word from memory
```

```
sw $t0, 8($s3)    #store word to memory
```

- ❑ The data is loaded into (lw) or stored from (sw) a register in the register file
- ❑ The memory address is formed by adding the contents of the **base address register** to the **offset** value
- ❑ Offset can be negative, and must be multiple of 4

MIPS Instruction set: Load Instruction

❑ Load/Store Instruction Format:

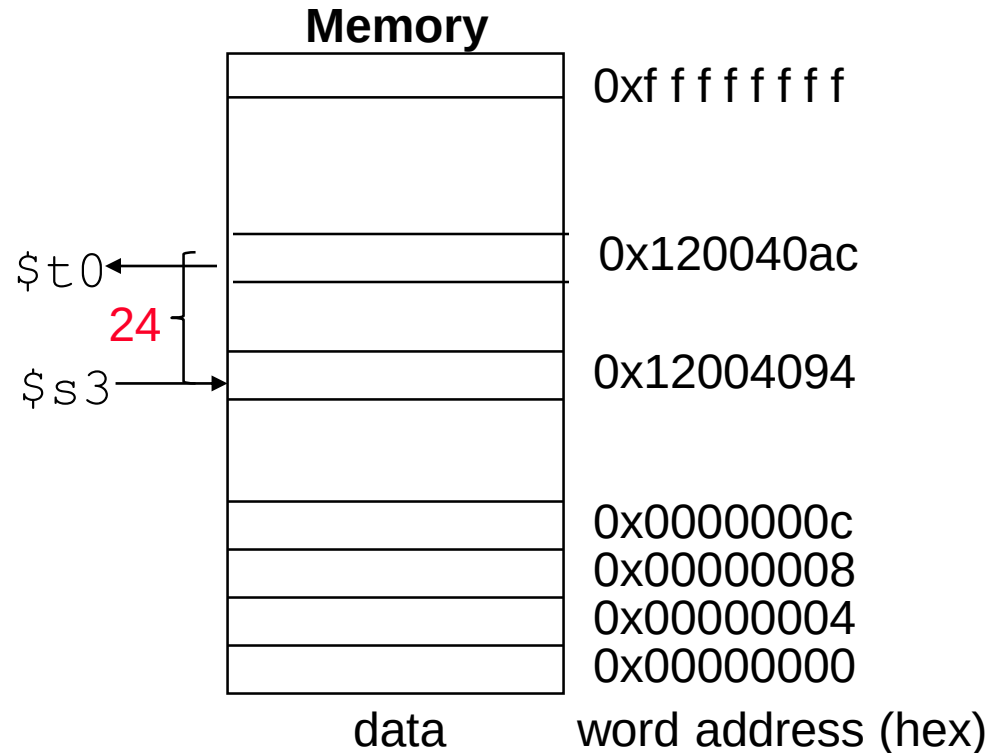
lw \$t0, 24(\$s3) # \$t0 ← mem at 24+\$s3

(move a word from memory to \$t0)

$$24_{10} + \$s3 =$$

$$\begin{array}{r} . \quad 0001 \quad 1000 \quad (24) \\ + \quad . \quad 1001 \quad 0100 \quad (94) \\ \hline . \quad 1010 \quad 1100 \quad (ac) \\ = \quad 0x1200 \quad 40ac \end{array}$$

$$0x..94 = ..1001 \ 0100$$



MIPS Control Flow Instructions

❑ MIPS conditional branch instructions:

```
bne $s0, $s1, Exit #go to Exit if $s0≠$s1
beq $s0, $s1, Exit #go to Exit if $s0=$s1
```

```
| Ex:      if (i==j)
           h = i + j;
```

```
           bne $s0, $s1, Exit
           add $s3, $s0, $s1
Exit :     ...
```

Example

start:

addi s0, zero, 2 #load value for s0

addi s1, zero, 2

addi s3, zero, 0

beq s0, s1, Exit

add s3, s2, s1

Exit: add s2, s3, s1

.end start

What is final value of s2?

In Support of Branch Instructions

- ❑ How to use `beq`, `bne`, to support other kinds of branches (e.g., branch-if-less-than)?
- ❑ Set flag based on condition: `slt`
- ❑ Set on less than instruction:

```
slt $t0, $s0, $s1    # if $s0 < $s1      then
                     # $t0 = 1             else
                     # $t0 = 0
```

- ❑ Alternate versions of `slt`

```
slti $t0, $s0, 25     # if $s0 < 25 then $t0=1 ...
sltu $t0, $s0, $s1    # if $s0 < $s1 then $t0=1 ...
sltiu $t0, $s0, 25    # if $s0 < 25 then $t0=1 ...
```

- ❑ How about set on bigger than?

Unconditional branch

- ❑ MIPS also has an unconditional branch instruction or **jump** instruction:

```
j    label           #go to label
```

Example

- ❑ Write assembly code to do the following

```
if (i<5)
    X = 3;
else
    X = 10;
```

Solution

```
    slti $t0,$s1,5      # i<5? (inverse condition)
    beq  $t0,$zero,else # if i>=5 goto else part
    addi $t1,$zero,3    # X = 3
    j    endif          # skip the else part
else: addi $t1,$zero,10  # X = 10
endif:...
```

Representation of MIPS instruction

- ❑ All MIPS instructions are 32 bits wide
- ❑ Instructions are 32 bits binary number

3 Instruction Formats: **all 32 bits wide**

op	rs	rt	rd	sa	funct	R format
op	rs	rt	immediate			I format
op	jump target					J format

Reference: MIPS Instruction Reference (MIPS_IR.pdf)

R-format instruction

- ❑ All fields are encoded by mnemonic names

op	rs	rt	rd	shamt	funct
----	----	----	----	-------	-------

op	6-bits	opcode that specifies the operation
rs	5-bits	register file address of the first source operand
rt	5-bits	register file address of the second source operand
rd	5-bits	register file address of the result's destination
shamt	5-bits	shift amount (for shift instructions)
funct	6-bits	function code augmenting the opcode

Example of R-format instruction

add \$t0, \$s1, \$s2

sub \$t0, \$s1, \$s2

- ❑ Each instruction performs **one** operation
- ❑ Each specifies exactly **three** operands that are all contained in the datapath's register file (\$t0, \$s1, \$s2)

destination ← source1 **op** source2

- ❑ Binary code of Instruction



Example

- ❑ Find machine codes of the following instructions

```
lw      $t0, 0($s1)    # initialize maximum to A[0]
addi    $t1, $zero, 0  # initialize index i to 0
add     $t1, $t1, 1     # increment index i by 1
```

Example of I-format instruction

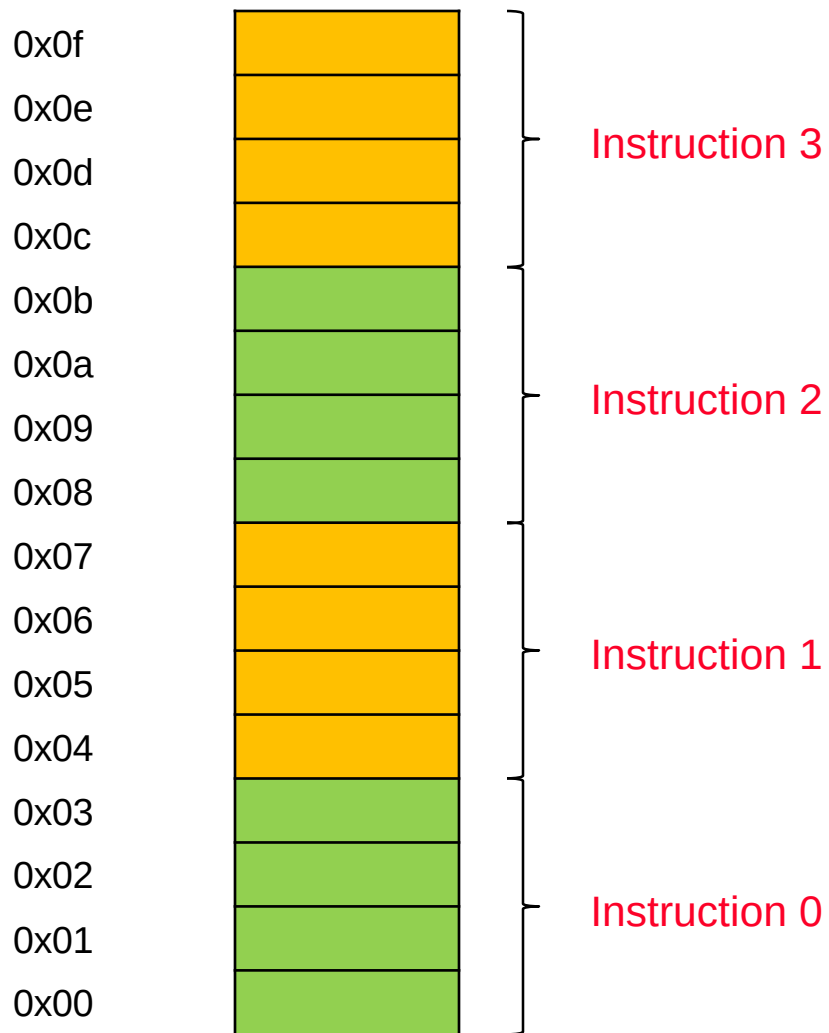
`slti $t0, $s2, 15` `#$t0 = 1 if $s2 < 15`

- ❑ Machine format (I format):

0x0A	18	8	0x0F
------	----	---	------

- ❑ The constant is kept **inside** the instruction itself!
 - ❑ Immediate format **limits** values to the range $+2^{15}-1$ to -2^{15}

How MIPS executes program?



❑ Stored program in memory

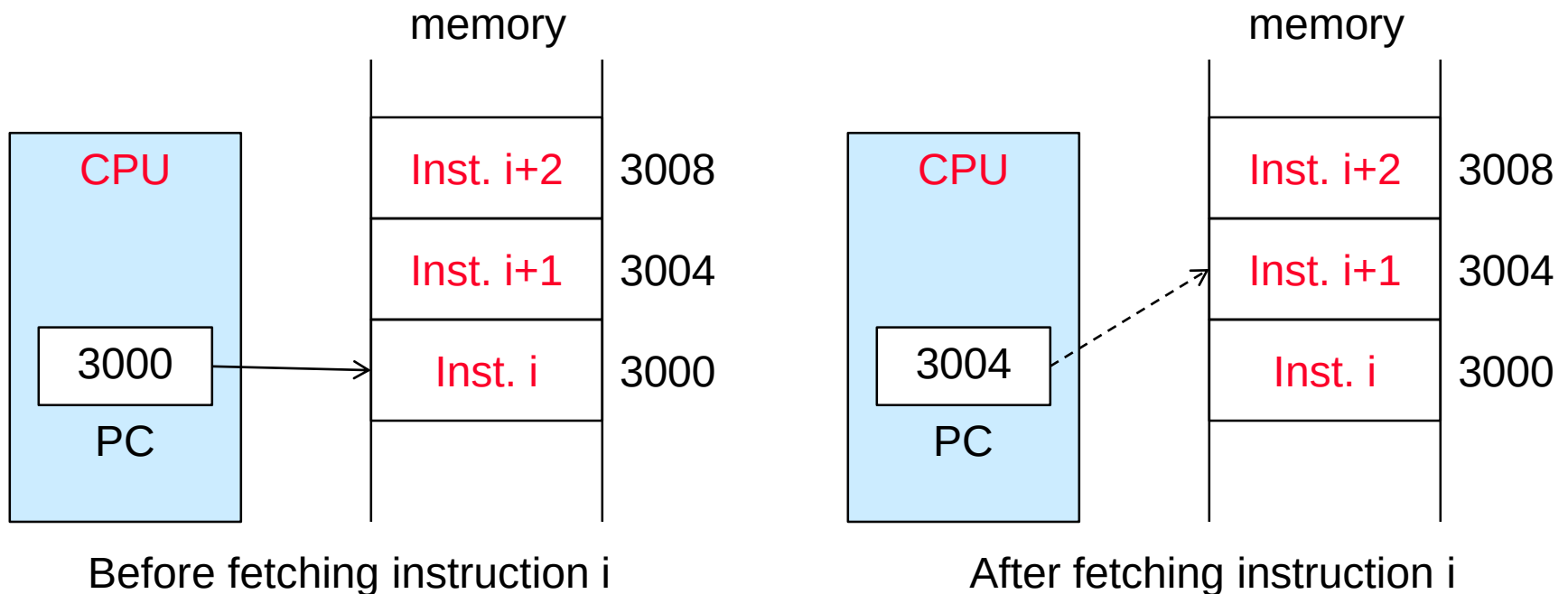
❑ 1 instruction = 1 word

❑ Instruction cycle

- ❑ Fetch
- ❑ Decode
- ❑ Execution

MIPS instruction cycle

- ❑ Program flow controlled by the PC register
 - ❑ **PC: Program Counter**
 - ❑ Specifies address of the next instruction to be fetched
 - ❑ Auto-increment after each instruction fetch
- ❑ MIPS instruction fetch cycle



Example

The simple while loop: `while (A[i]==k) i=i+1;`

Assuming that: `i`, `A`, `k` are stored in `$s1`, `$s2`, `$s3`

Solution

```
loop: add    $t1,$s1,$s1        # t1 = 4*i
      add    $t1,$t1,$t1        #
      add    $t1,$t1,$s2        # t1 = A + 4*I,
                                # address of A[i]
      lw     $t0,0($t1)         # load data in A[i]
                                # into t0
      bne    $t0,$s3,endwhl     #
      addi   $s1,$s1,1          #
      j      loop              #
endwhl: ...                     #
```

Example

The simple switch

```
switch(test) {  
    case 0:  
        a=a+1; break;  
    case 1:  
        a=a-1; break;  
    case 2:  
        b=2*b; break;  
    default:  
  
}
```

Assuming that: test, a, b are
stored in \$s1, \$s2, \$s3

Solution

```
        beq    s1,t0,case_0  
        beq    s1,t1,case_1  
        beq    s1,t2,case_2  
        b      default  
case_0:  
        addi   s2,s2,1      #a=a+1  
        b      continue  
case_1:  
        sub    s2,s2,t1     #a=a-1  
        b      continue  
case_2:  
        add    s3,s3,s3     #b=2*b  
        b      continue  
default:  
continue:
```

Example

- ❑ Write assembly code correspond to the following C code

```
for (i = 0; i < n; i++)  
    sum = sum + A[i];
```

loop:

```
add    s1,s1,1        #i=i+step  
add    t1,s1,s1        #t1=2*s1  
add    t1,t1,t1        #t1=4*s1  
add    t1,t1,s2        #t1 <- address of A[i]  
lw     t0,0(t1)        #load value of A[i] in t0  
add    s5,s5,t0        #sum = sum+A[i]  
bne    s1,s3,loop      #if i != n, goto loop
```

Exercise

❑ How branch instruction is executed?

```
slti $t0,$s1,5
```

```
bne $t0,$zero,else →
```

**How can CPU jump from here
to the “else” label?**

```
addi $t1,$zero,3
```

```
j     endif
```

```
else: addi $t1,$zero,10
```

```
endif:...
```

Instructions for Accessing Procedures

- ❑ MIPS **procedure call** instruction:

`jal ProcedureAddress #jump and link`

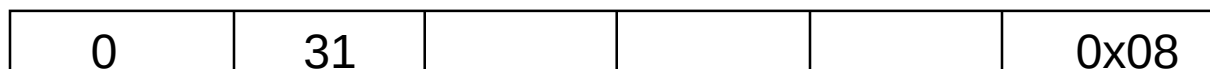
- ❑ Saves PC+4 in register \$ra to have a link to the next instruction for the procedure return
- ❑ Machine format (**J** format):



- ❑ Then can do procedure **return** with

`jr $ra #return`

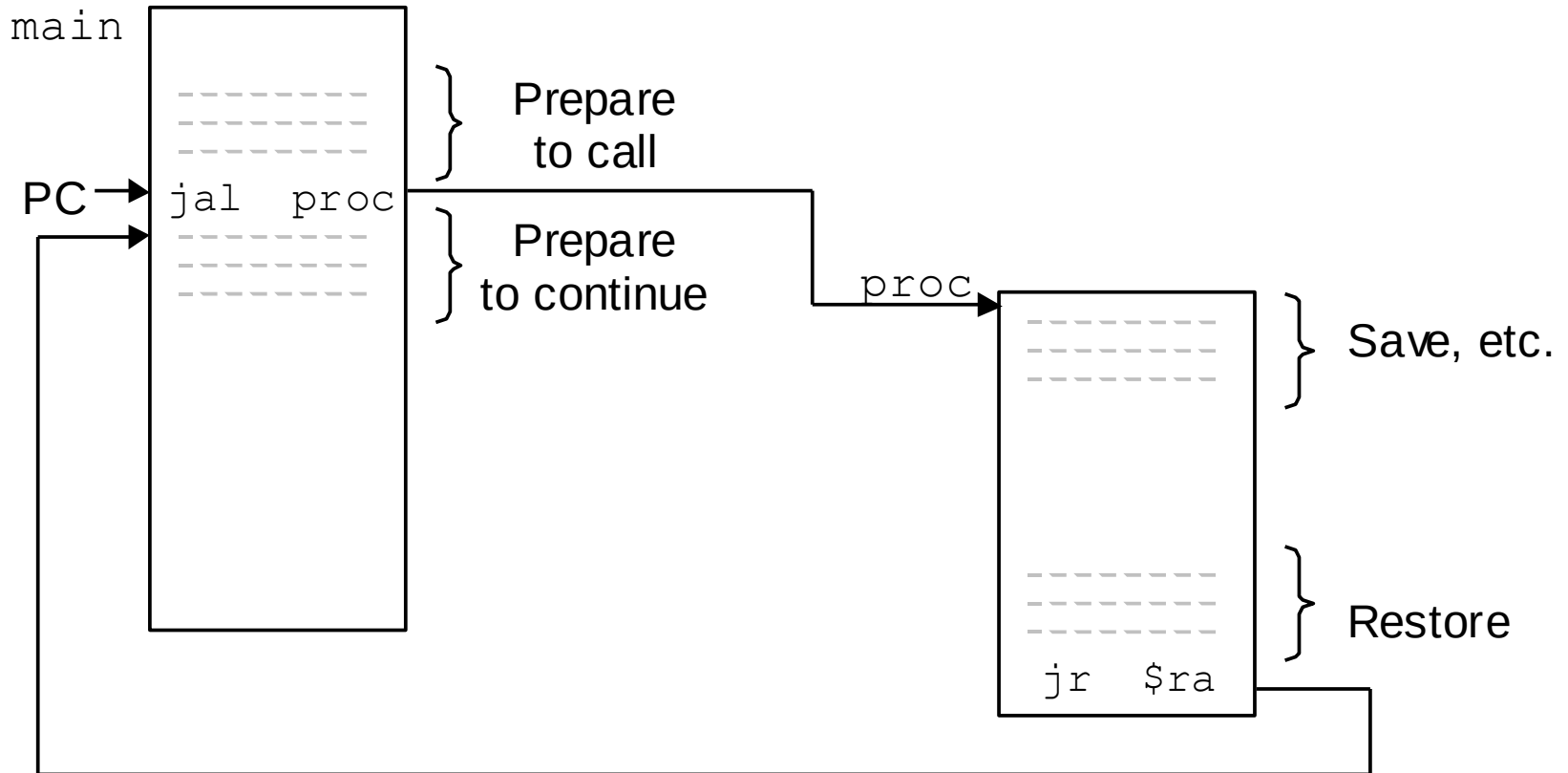
- ❑ Instruction format (**R** format):



Six Steps in the Execution of a Procedure

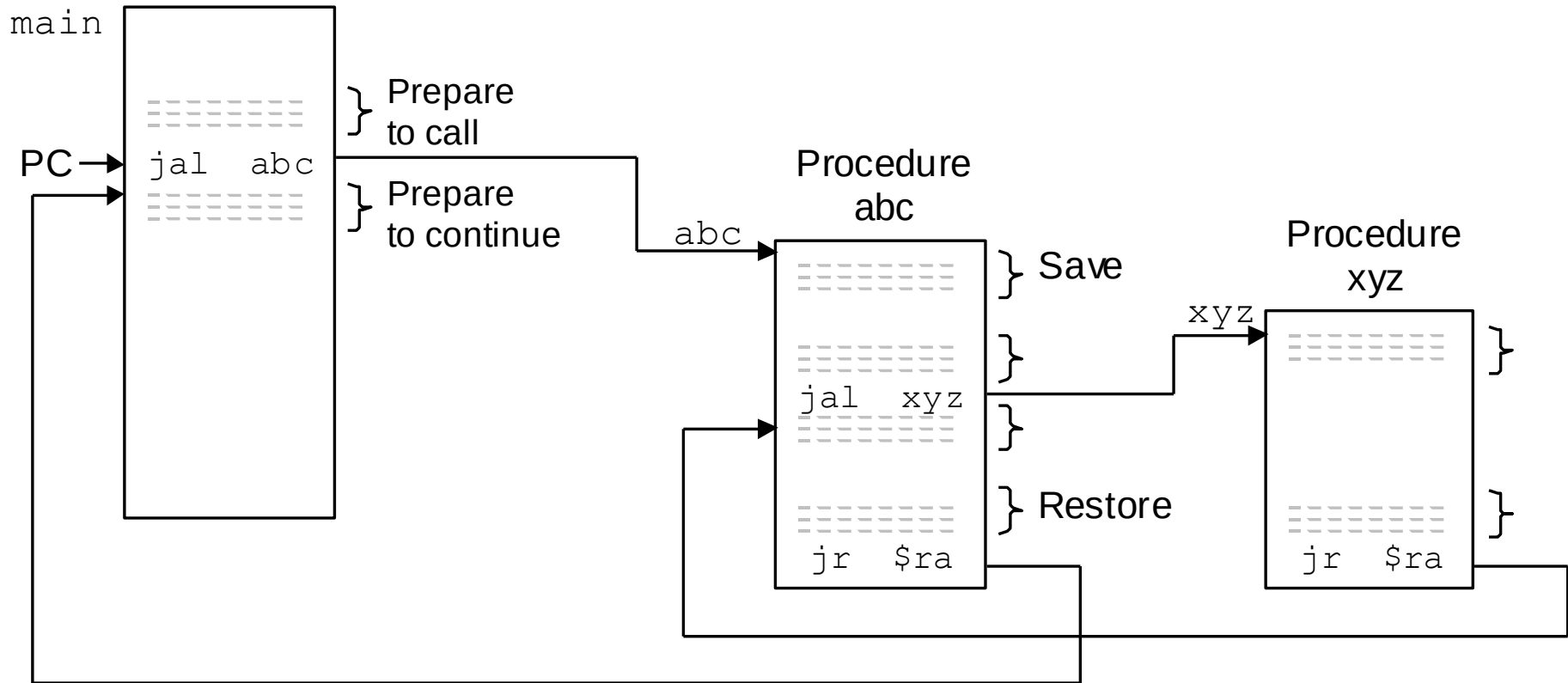
1. Main routine (**caller**) places parameters in a place where the procedure (**callee**) can access them
 - | `$a0 - $a3`: four **argument** registers
2. **Caller** transfers control to the **callee** (**jal**)
3. **Callee** acquires the storage resources needed
4. **Callee** performs the desired task
5. **Callee** places the result value in a place where the **caller** can access it
 - | `$v0 - $v1`: two **value** registers for result values
6. **Callee** returns control to the **caller** (**jr**)
 - | `$ra`: one **return address** register to return to the point of origin

Illustrating a Procedure Call



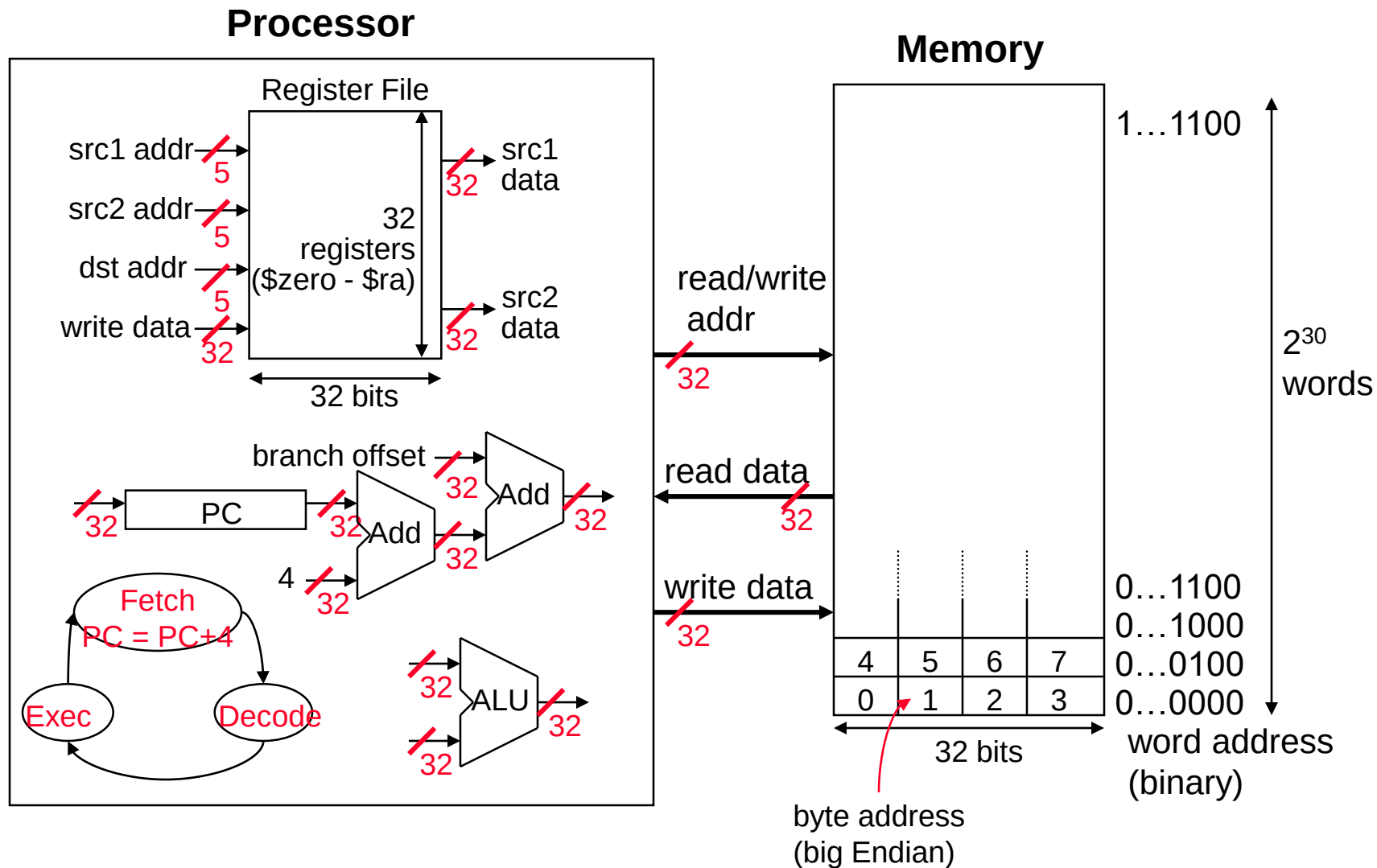
Relationship between the main program and a procedure.

Nested Procedure Calls



Example of nested procedure calls.

MIPS Organization



Summary

- ❑ Provided one problem to be solved by computer
 - | Can it be implemented?
 - | Can it be programmed?
 - | Which CPU is suitable?
 - ❑ Metric of performance
 - | How many bytes does the program occupy in memory?
 - | How many instructions are executed?
 - | How many clocks are required per instruction?
 - | How much time is required to execute the program?
- ➔ Largely depend on Instruction Set Architecture (ISA)